Low-Power D-Flip-Flop and Adaptive Clocking via GDI–CMOS Integration

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***Abstract*—In this project we designed a low-power sequential circuit. This circuit mainly consists of a D-flip-flop and the essential blocks for an adaptive clocking scheme. This system is hybrid where the d flip flop build with GDI technique, while the adaptive clocking construct based on CMOS technology. This hybrid methodology was chosen to optimize overall performance, specially to attenuate issues such as static power consumption (leakage) across various operational states, which becomes particularly noticeable at smaller process technologies.**

After many attempts, exploring multiple designs, and fine-tuning various parameters, we successfully achieved a theoretical D-flip-flop design containing only 12 transistors, demonstrating efficient area and power utilization. However, ongoing simulation and optimization efforts are still required to achieve the desired clean waveform characteristics and robust performance in practice, as confirmed by the current simulation results.

***Index Terms*—GDI, D-flip flop, 12 transistor, 22nm, Ring-oscillator, Phase Detector, VCO, CMOS, Low-power design, Adaptive clocking, Leakage reduction.**

I. Introduction

In modern System-on-Chip (SoC) designs, low-power memory units is a critical need, particularly flip-flops (FFs). To address this, our proposed system design features of an adaptive clocking. However, despite that this design approach is promising, the implementation of this system at advanced nodes introduces significant timing challenges, including clock jitter (frequency oscillation), sensitivity to Process-Voltage-Temperature (PVT) variations, and layout complexities like optimizing for shared diffusion to minimize area. Achieving the desired robust performance and clean waveform characteristics, particularly concerning the stability and noise immunity of the flip-flop output, has proven to be an ongoing challenge in preliminary simulations.

A primary limitation of traditional CMOS logic in flip-flop design is its relatively higher power consumption compared to alternative approaches. Furthermore, conventional open-loop adaptive clock generators often fail to adequately compensate for PVT variations and struggle to maintain the required timing precision.

Therefore, our proposed solution employs a dual-approach design methodology, combining a GDI-based D-flip-flop with a CMOS-based Phase-Locked Loop (PLL) for adaptive clock generation. When GDI technology applyed to the flip-flop design it significantly reduce transistor count, resulting to a more compact layout, minimized parasitic capacitances, and hence having a lower power consumption memory module. Conversely, CMOS technology is utilized for the adaptive clocking unit to achieve a robust and stable clock generation that effectively tracks and compensates for environmental variations.

Ultimately, the primary objective of this project is to achieve significant enhancements in both area efficiency and power consumption for sequential circuits in SoC designs.

II. Background and Related Work

A. Principle of GDI

GDI (Gate Diffusion Input) - one technique of low power digital circuit design. Which allows reducing power consumption, delay and area of digital circuits, where the logic design maintains with low complexity. Comparing GDI to traditional CMOS and other design techniques is presented, with respect to the layout area, number of devices, delay and power dissipation, illustrating the strengths and weaknesses of the GDI method compared to other ones. Many basic logic gates have been implemented in 22 nm technology to compare the GDI technique with CMOS. In addition to, there were a prototype test chip of 8-bit CLA adder has been fabricated, based on GDI and CMOS cell libraries, showing up to 45 % reduction in power-delay product in GDI.

B. CMOS vs GDI

The key differences between Gate Diffusion Input (GDI) and standard CMOS logic circuits are as shown on Table 1:

TABLE I
Comparison Between GDI and Standard CMOS Logic

GDI introduces the idea of reduction transistor count and area, presenting lower power and faster operation, but suffers non-full-swing output, requires twin-well/SOI processes, and has VT-drop issues; While CMOS provides full-swing outputs, standard bulk connections, robust signal integrity, but at higher area and power cost.

C. Importance of PLL

A phase-locked loop (PLL) is a closed-loop control circuit that adjusts a voltage-controlled oscillator (VCO) to track the frequency and phase of a reference signal. By converting timing errors into UP/DOWN pulses (via the phase detector), then into an analog control voltage (via the charge pump and loop filter), it mitigates for process, voltage, and temperature variations and suppresses jitter. A programmable frequency divider feeds back the VCO output to the detector, completing the negative-feedback loop that achieves both frequency and phase lock.

D. Definitions Key Concepts

GDI Cell. This cell consists of one NMOS and one PMOS, where gate inputs share a common net and source/drain nodes serve as the logic inputs. GDI significantly cuts transistor count (and hence the capacitance) as compare to standard CMOS.

CMOS Inverter. A pair pull-up (PMOS) and pull-down (NMOS) network. Where transistors implementing the Boolean NOT function, which produce full rail-to-rail swings and good noise margins.

Transmission Gate. A bidirectional pass-gate made from NMOS and PMOS connected in parallel, used to pass signals without the VT-drop penalty that occurs at a single pass transistor.

Master–Slave D-Flip-Flop. Two level-sensitive latches (master, slave) clocked in opposite phases to end up with a storage element of its D input only at a clock transition.

Phase Detector (PD). A circuit Compares the phase of two clock signals and then produces “UP” or “DOWN” control pulses whose width encodes their timing difference. In other words, it compares each rising edge of the reference clock with the feedback clock: when the reference leads, it issues an UP pulse, and when it lags, a DOWN pulse—each pulse’s width proportional to the timing difference. These pulses drive the charge pump and loop filter, generating a control voltage that adjusts the VCO until lock.

XOR Phase Detector. A simple combinational PD that outputs HIGH whenever two inputs differ; its pulse duration is proportional to phase error but only works over ±180°.

Charge Pump. An analog block converting UP/DOWN logic pulses into currents using PMOS and NMOS switches that exactly charge or discharge the loop-filter capacitor, generating a smooth control voltage for adjusting VCO.

III. System Architecture

A. Modified GDI Master–Slave D-Flip-Flop

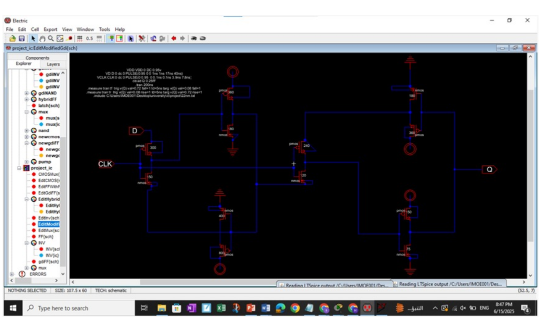

Fig. 1. Transistor Usage in the Modified GDI D–Flip–Flop

This design implements an edge-triggered D–FF using two GDI transmission-gate latches (master and slave) and four GDI inverters as shown in Figures below, for a total of 12 transistors:

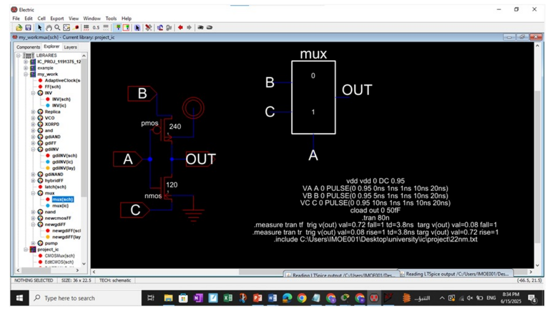

Fig. 2. Modified GDI mux

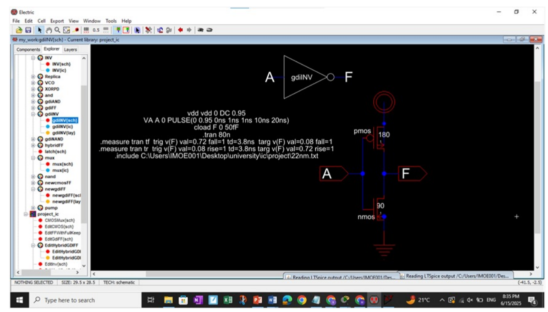

Fig. 3. Modified GDI inverter

On the rising edge of CLK, the master GDI MUX transparently samples D; on the falling edge, the slave GDI MUX drives the stored bit through three successive GDI inverters (two in-path stages plus a final buffer) to the output Q. This modified GDI architecture relies on only 12 transistors—over 40 % fewer than a conventional CMOS D–FF—and hence it absolutely reducing silicon area and parasitic capacitance. The lighter load enables faster clocking and shorter setup/hold times, while fewer switching nodes cut dynamic and short-circuit power.

Moreover, device count and speed, GDI offers strong signal integrity: its retrieve fully restore any threshold-voltage loss caused by the single-transistor pass-gates, guaranteeing true 0 V and V_{DD} swings at Q. Moreover, all pass-transistors employ native body ties (PMOS bodies to V_{DD}, NMOS bodies to GND), preventing latch-up and stabilizing threshold voltages which is very important for maintaining reliable logic levels and optimizing variability across process, voltage, and temperature corners. This combination of compactness, speed, power efficiency, and robust voltage margins makes the modified GDI D–FF exceptionally well-suited for high-performance, low-power designs in advanced 22 nm technologies.

B. XOR Phase Detector

(see Fig. 4)

An XOR–based phase detector is the simplest combinational circuit for measuring phase difference between two periodic signals. It drives its output HIGH whenever its inputs A and

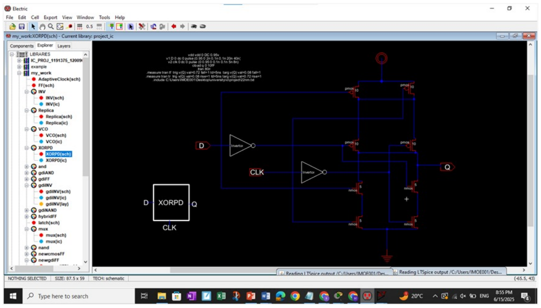

Fig. 4. XOR-based phase detector block diagram.

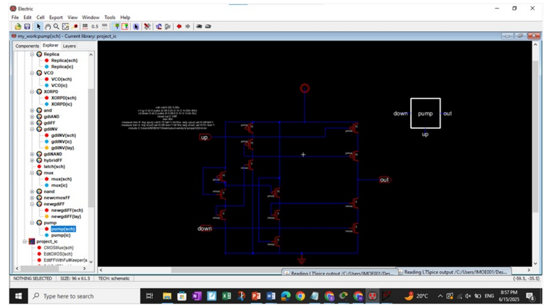

Fig. 6. Charge-pump block diagram.

B differ, and LOW when they are equal. As a result, the duty cycle of the output pulse train is directly proportional to the phase error between A and B. However, this detector can only measure phase differences up to $\pm 180^\circ$ and provides no inherent memory (i.e., flip-flops or reset circuitry are not required).

C. Ring VCO

(see Fig. 5)

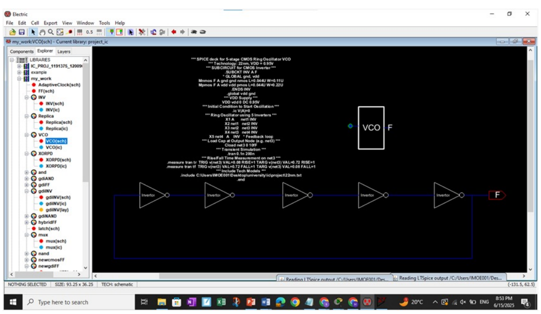

Fig. 5. Ring VCO block diagram.

The VCO is realized as a five-stage static CMOS ring oscillator: five inverters in series feed their output back to the input, yielding an odd number of inversions (180° total phase shift) that sustains oscillation. Each inverter contributes a propagation delay t_{pd}, so the oscillation period is:

$$T = 2N\, t_{pd} \quad (N = 5)$$

and the free-running frequency is:

$$f_{\text{osc}} = \frac{1}{2N\, t_{pd}}.$$

Node "A" is the loop's input (also used for biasing in a current-starved variant), and "F" is the oscillation output. This topology is compact and easy to integrate in CMOS, though it trades off higher phase noise and limited frequency stability.

D. Charge Pump

(see Fig. 6)

The charge pump is built from matched PMOS current sources and NMOS current sinks gated by the UP and DOWN control signals. When UP pulses high, its PMOS source devices inject a constant current into the loop-filter node; when DOWN pulses high, its NMOS sink devices draw the same current out. A replica branch—sized identically to the main pump—feeds a small test capacitor and drives an error amplifier to adjust the bias voltage so that UP and DOWN currents match precisely. Matched device sizing and the replica feedback loop minimize current-mismatch spurs and ensure linear charge-transfer, yielding a low-jitter, high-linearity PLL core.

IV. Simulation Results

A. Waveforms

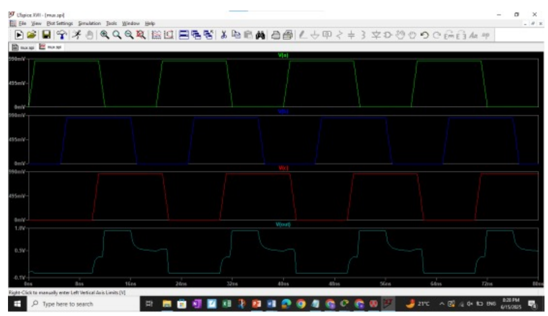

Fig. 7. Waveform of Modified GDI Mux.

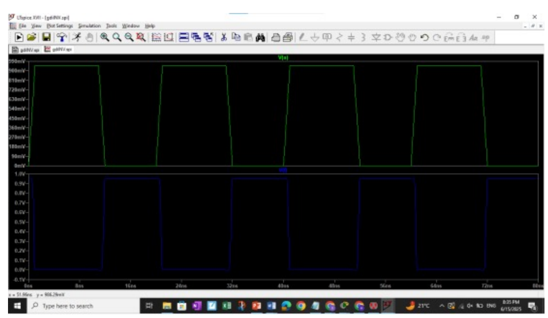

Fig. 8. Waveform of Modified GDI Invertor.

B. Specifications

- **Technology Node:** 22 nm (0.022 μm)
- **Transistor Details:**
  - **NMOS Transistors** (6 devices):

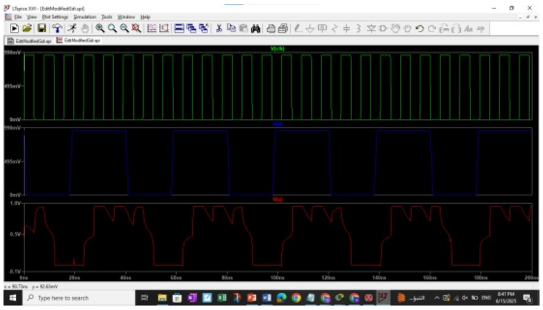

Fig. 9. Waveform of D-flip flop.

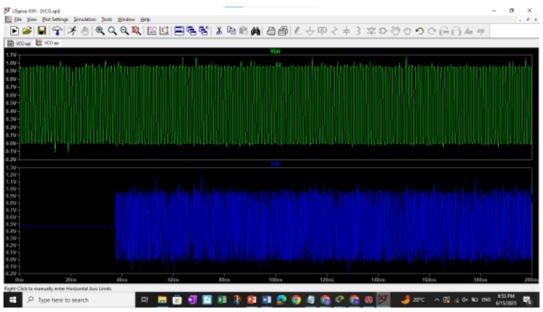

Fig. 10. Waveform of VCO.

- ∗ GDI master MUX: $L \times 150W$
- ∗ Feedback inverter (master): $L \times 180W$
- ∗ Master-stage inverter: $L \times 400W$
- ∗ GDI slave MUX: $L \times 120W$
- ∗ Feedback inverter (slave): $L \times 180W$
- ∗ Slave-stage inverter: $L \times 75W$

– **PMOS Transistors** (6 devices):
- ∗ GDI master MUX: $L \times 300W$
- ∗ Feedback inverter (master): $L \times 360W$
- ∗ Master-stage inverter: $L \times 800W$
- ∗ Feedback inverter (slave): $L \times 360W$
- ∗ GDI slave MUX: $L \times 240W$
- ∗ Slave-stage inverter: $L \times 150W$

C. Transistor Area Calculations (22 nm)

a)

1) **NMOS Transistors**

$$L = 1 \times 22\,\text{nm} = 22\,\text{nm} = 0.022\,\mu\text{m},$$

$$\begin{aligned} A_{\text{NMOS, total}} &= 6 \times A_{\text{per device}} \\ &= 0.04862\,\mu\text{m}^2 = 48.62\,\text{nm}^2. \end{aligned}$$

2) **PMOS Transistors**

$$L = 1 \times 22\,\text{nm} = 22\,\text{nm} = 0.022\,\mu\text{m},$$

$$\begin{aligned} A_{\text{PMOS, total}} &= 6 \times A_{\text{per device}} \\ &= 0.02431\,\mu\text{m}^2 = 24.31\,\text{nm}^2. \end{aligned}$$

3) **Total Area**

$$\begin{aligned} A_{\text{total}} &= A_{\text{NMOS, total}} + A_{\text{PMOS, total}} \\ &= 0.04862\,\mu\text{m}^2 + 0.02431\,\mu\text{m}^2 \\ &= 0.07293\,\mu\text{m}^2 = 72.93\,\text{nm}^2. \end{aligned}$$

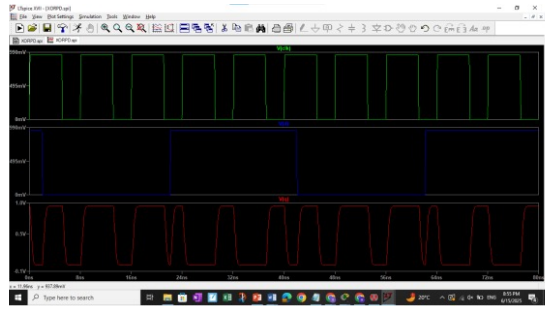

Fig. 11. Waveform of XORPD.

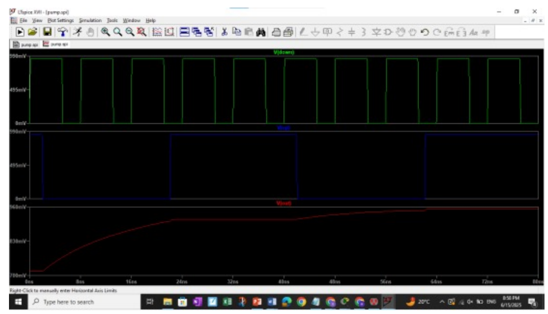

Fig. 12. Waveform of Charge-pump.

D. Delay and Power

a)

1) **Total Area**

$$\begin{aligned} A_{\text{total}} &= A_{\text{NMOS, total}} + A_{\text{PMOS, total}} \\ &= 0.04862\,\mu\text{m}^2 + 0.02431\,\mu\text{m}^2 \\ &= 0.07293\,\mu\text{m}^2 = 72.93\,\text{nm}^2. \end{aligned}$$

Dynamic Power:

$$\begin{aligned} P_{\text{dyn}} &= \tfrac{1}{2}\,C_{\text{load}}\,V_{dd}^2\,f \\ &= 0.5 \times 25\,\text{fF} \times (0.95\,\text{V})^2 \times \tfrac{1}{7.8\,\text{ns}} \\ &\approx 1.45\,\mu\text{W}. \end{aligned}$$

V. Discussion

This project introduced a hybrid low-power design consisting of a GDI-based D-flip-flop and a CMOS-based adaptive clock generator. The flip-flop design was successfully reduced to just 12 transistors, leading to significant savings in silicon area and dynamic power consumption, while ensuring strong signal integrity and full voltage swing.

The integration of a replica charge pump in the PLL loop helped achieve better current matching, reducing jitter and improving timing accuracy under PVT (Process, Voltage, Temperature) variations. Overall, the design demonstrates a promising balance between performance, efficiency, and area optimization—particularly suited for advanced 22nm technologies.

Improvement Suggestions: Decreasing the rise time can be achieved by increasing the frequency of the clock signal.

Poor logic '1' level at the output can be mitigated by increasing the width of the master's feedback inverter.

To eliminate poor logic '1', the width of the slave's feedback inverter can be decreased.

Extending the duration of a good logic '1' at the output can be done by increasing the clock duty cycle.

Reducing the rise time can also be further enhanced by boosting the clock frequency.

Additional improvements may include implementing calibration techniques, replacing the ring VCO with an LC-VCO for better spectral purity, and laying out the full design to analyze parasitics and area overhead in physical implementation.

VI. Conclusion

In this project, we presented a hybrid low-power sequential circuit that integrates a GDI-based D-flip-flop and a CMOS-based adaptive clock generator. The primary objective was to reduce area and power consumption while ensuring robust timing performance across varying PVT (process, voltage, temperature) conditions.

Through the use of Gate Diffusion Input (GDI) logic, we successfully designed a compact 12-transistor flip-flop that offers significant reductions in transistor count and dynamic power consumption. This minimalistic design effectively addresses the challenges of layout area and parasitic capacitance, which are critical concerns in advanced 22nm technologies. At the same time, signal integrity was maintained due to native body ties and full voltage swing restoration, ensuring functional reliability.

To complement the GDI-based flip-flop, we implemented an adaptive clock generator using standard CMOS techniques. This block integrates a phase-locked loop (PLL) architecture enhanced with a replica charge pump, enabling precise current matching and improving frequency stability. The system addresses jitter, leakage, and timing variability by adapting to operational states, offering a more energy-efficient solution for modern SoC designs.

Although simulation results show promising improvements in terms of power and area efficiency, additional fine-tuning is needed to further refine waveform quality and enhance robustness under practical conditions. Nevertheless, this work lays a solid foundation for future exploration and prototyping of ultra-low power sequential logic systems using hybrid design techniques.

VII. References

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